

Revision Notes for Class 12 Computer Science Chapter 9 – Structured Query Language (SQL)

Structured Query Language (SQL) is an essential topic for Class 12 Computer Science, enabling students to work efficiently with databases. SQL is used to create, modify, and operate databases in relational database management systems (RDBMS) such as MySQL, Oracle, and Microsoft SQL Server. It stands out because its commands are simple, use English words, and are not case sensitive.

Instead of writing complex application programs for data access, SQL lets users simply specify what information they want from the database. This separation of “what” from “how” makes SQL both powerful and user-friendly for database operations. Understanding how SQL functions is crucial for further studies and real-world database applications.

Basics of SQL and MySQL Setup

SQL allows users to perform a variety of tasks such as creating databases, populating them with data, and retrieving results using queries. To get started, students should install MySQL from the official website and start the MySQL service. Once the `mysql>` prompt appears, MySQL is ready for use. Always remember that SQL statements must end with a semicolon.

- SQL is case insensitive: Salary and SALARY are treated the same.
- To write multiple-line statements, omit the semicolon until the last line, and the prompt will indicate continuation.

Data Types and Constraints

Every attribute or column in a table has a specific data type that controls what kind of data can be stored. The most commonly used data types in MySQL are CHAR(n) for

fixed-length strings, VARCHAR(n) for variable-length strings, INT for integers, FLOAT for decimal numbers, and DATE for dates.

Constraints are rules applied to columns for maintaining data integrity. The main types of constraints used in SQL are:

- NOT NULL – prevents missing or unknown values.
- UNIQUE – ensures all values are distinct in a column.
- DEFAULT – provides a value if none is supplied during insertion.
- PRIMARY KEY – one or more columns uniquely identify each record.
- FOREIGN KEY – enforces the relationship between tables, usually referencing a PRIMARY KEY in another table.

Creating and Modifying Tables

Tables are the core structure in a relational database. Each table has columns (attributes) with specified data types and optional constraints. Creating a database uses the command CREATE DATABASE databasename; followed by USE databasename; to operate within it. To list all tables in the selected database, SHOW TABLES; is used.

To define a table, the CREATE TABLE statement is used. For example, in the STUDENT table, typical columns are RollNumber (INT, PRIMARY KEY), SName (VARCHAR(20), NOT NULL), SDateofBirth (DATE, NOT NULL), and GUID (CHAR(12), FOREIGN KEY). Other tables such as GUARDIAN and ATTENDANCE are similarly structured, with their own attributes and constraints.

- ALTER TABLE is used to change the structure, such as adding or dropping columns, modifying data types, or changing constraints.

- To add a composite primary key, specify multiple columns in the PRIMARY KEY clause.
- DROP TABLE and DROP DATABASE are permanent actions and cannot be undone.

Data Manipulation: Insert, Update, Delete

Data in tables can be inserted using INSERT INTO tablename VALUES(...) or by specifying column names. When working with tables that have foreign key constraints, ensure the referenced table already contains required data, otherwise insertion will fail.

To change data, the UPDATE statement is used, and for removing data, the DELETE statement. Always include a WHERE clause when updating or deleting records to prevent altering all rows unintentionally.

- SELECT * FROM tablename; is used to display all records of a table.
- UPDATE tablename SET column = value WHERE condition; is for modifying values conditionally.
- DELETE FROM tablename WHERE condition; is for deleting rows conditionally.

Querying Data and Filtering Results

The SELECT statement is fundamental for retrieving data. You can select specific columns or all columns (using *), and use the WHERE clause to filter results based on conditions. The query can be enhanced using operators like IN, LIKE, BETWEEN, and logical operators.

ORDER BY arranges output in ascending or descending order. The DISTINCT keyword ensures only unique values are returned in the result. You may use pattern matching with LIKE and wildcards % and _, for example, Ename LIKE 'K%' finds all names starting with 'K'.

SQL Functions: Single Row and Aggregate

SQL offers a variety of functions. Single row (scalar) functions process individual rows, such as mathematical (POWER, ROUND, MOD), string (UPPER, LOWER, LENGTH, LEFT, MID, RIGHT, INSTR, TRIM), and date/time (NOW, DATE, MONTHNAME, YEAR, DAY, DAYNAME) functions.

Aggregate (multiple row) functions operate over groups of rows and include MAX, MIN, AVG, SUM, COUNT(*), and COUNT(column). They are especially useful for summarizing data and are often used with the GROUP BY clause to aggregate results by a specific column or set of columns.

For example:

- `SELECT MAX(Price) FROM INVENTORY;` finds the highest price.
- `SELECT COUNT(DISTINCT Model) FROM INVENTORY;` counts unique models.

Grouping, Joining, and Set Operations

The GROUP BY clause clusters rows that have the same values in specified columns, allowing aggregate functions to be computed in each group. HAVING is used to filter these groups based on aggregate results.

SQL supports set operations like UNION, INTERSECT, and MINUS to combine results from multiple queries. UNION combines all unique results; INTERSECT finds common rows, while MINUS returns those present in one result but not another.

Cartesian product (cross join) creates all possible pairings between two tables, while JOIN is used to merge related rows from multiple tables based on matching values of specified columns.

There are various ways to join tables, such as using the WHERE clause, explicit JOIN ON clause, or NATURAL JOIN, which merges tables automatically on columns with the same name, removing duplicates. In general, to join N tables, N-1 joins are needed.

Key Differences and Definitions

Understanding differences between SQL statements is crucial:

- ALTER changes table structure, while UPDATE modifies data within the table.
- DELETE removes data but keeps the table, whereas DROP deletes the entire table structure.

A Relational Database Management System (RDBMS) is a software system for managing databases that stores data in tables with relationships among them. MySQL and Oracle are widely used examples of RDBMS software.

Common Exam-Focused Command Snippets

Students should practice frequently used SQL snippets for exams:

- `SELECT SName, SDateofBirth FROM STUDENT WHERE RollNumber=1;`
- `UPDATE tablename SET column1 = value1 WHERE condition;`
- `DELETE FROM tablename WHERE condition;`
- `SELECT COUNT(*) FROM tablename;`

By regularly working with these commands and understanding SQL operations, constraints, and data types, students can build a strong foundation for performing efficient data management and analytical tasks, which is essential for practical and theoretical computer science examinations.